

Class 23 Fields

Goals

1. Understand the definition of a field.
2. Be able to prove simple properties of fields.
3. Develop familiarity with examples.

Definition: A field is a nonempty set, F , with two binary operations called addition and multiplication satisfying 3 properties:

Examples

Real numbers - rational and irrational

We commonly think of decimal representation for a real number.

Rational numbers have decimal representations that repeat periodically. Irrational numbers have decimal representations that are nonrepeating.

Practice: Write each repeating decimal as a ratio of integers.

$$a. 0.\overline{123} = x$$

$$b. 7.2959595\overline{95} = x$$

Properties of fields

How do the two operations in a field interact? For example, 0 is the additive identity. How does it work with multiplication?

Similar Problem: If -1 is the additive inverse of the multiplicative identity in F , and x is any element of F , show that the product of -1 with x produces the additive inverse of x .

Homework:

Prove: If x is any element of a field F , then $-1 \cdot x = -x$.

Prove that $\mathbf{Z}_6$ is not a field.